Course Data Analytics

Assigment Research Assignment 1

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Student Imameleng Mpitso

Section A : Data fundomentals

Question 1

A database is an organised place where data is stored so it can be easily saved, updated and retrieved.
Database are importand because the allow companies to store large amounts of information safely and access it quickly.
Examples : bank customer records, school student records.

Question 2

A data warehouse stores large amounts of historical data from many sources for analysis.
Unlike a database it is mainly use for reporting, not daily transactions.
Companies use data warehouse to analyze trends and make decisions

Question 3

A data lake stores raw data in its original format (structured and unstructed)
unlike a data warehouse data is not cleaned before storage
Business prefer data lakes when working with big data, images, text or machine learning

Question 4

A dart mart is a smaller part of a data warehouse focused on one department.
It is usually used by teams like sales, finance or HR for quick analysis

Question 5

Structured : organised in tables (e.g Excel sales data)
Semi structured : has some structure (e.g JSON files)
Unstructured : no fixed format (e.g images)

Question 6

Data modeling is designing how data is stored and connected
It is important because it makes data systems efficient and easy to understand.
Types :
- Conceptual model
- Logical model

Question 7

Data mining is finding patterns and insights from large data sets.
Examples:
- Online store recommending products
- Banks detecting fraud

Question 8

A DBMS is a software that manages databases

DBMS	Description	use case
MySQL	open-source relational DB	small website
PostgreSQL	Advanced relational DB	Data analytics
Oracle	Enterprise DB	Large companies
MongoDB	NoSQL DB	Flexible data

Question 9

ETL
Extract : get data from sources
Transform : clean and format data
Load : store data in a warehouse

ETL is important for reliable and accurate analysis

Question 10

Data storage formats
CSV : simple tables easy to use
JSON : Flexible used in APIs
Parquet : Fast analytics big data
Avro : data streaming systems

Section B : AI and Machine Learning

Question 11

AI is when machine perform tasks that require human intelligence.

Narrow AI : designed for specific tasks (e.g voice assisstans)

General AI : human-like intelligence (~~does not yet exist~~ chatGPT)

Question 12

Machine Learning allows machines to learn from data instead of being programmed.

Supervised : data has labels

UnSupervised : finds patterns

Renforcement : learns by rewards

Question 13

Deep Learning is a type of Machine Learning using neural networks

Neural networks work like the human brain using layers to process data

Question 14

Natural Language Processing (NLP) helps computers understand human Languaga

Examples :

- spelling check
- chat bots
- voice assistants

Question 15

Large Language Model (LLM) understands and generates text

It is trained on massive text data

Example :

- chat GPT
- claude

Question 16

Generative AI creates new content like text or images

Tools :

- chat GPT – Text
- DALL·E – Images
- copilot – Coding and writing

Question 17

Training Datasets are used to teach an AI model
High quality data is important to avoid errors and bias
Bad data leads to incorrect or unfair results

Question 18

Computer vision allows machines to understand images and videos. Example : Facial recognition, medical image analysis

Section C : APIs and TOOLS

Question 19

API - Application Programming Interface lets software systems communicate. It is like a waiter taking orders between a customer and a kitchen.
API allow apps to share data

Question 20

Application Programming Interface (API) key is a secret code that allows access to an API
It must be protected.
If exposed others can misuse the service.

Question 21

REST - Representational State Transfer
GET - Retrieve data
POST - Create data
PUT - Update data
DELETE - Remove data

Question 22

JSON is a simple data format used for data exchange
JSON - JavaScript Object Notation

```
1 {
2   "name" : "Alex",
3   "age" : 25
4 }
```

Question 23

Prompt Engineering is writing clear instructions for AI

Poor - "Explain data"

Better - "Explain data analytics for beginners using simple examples"

Question 24

A map function applies the same operation to each element

```
1 numbers = [1, 2, 3]
2 squared = map ( Lambda x: x*x , Numbers)
```

Question 25

Cloud computing delivers service over the internet

- IaaS - virtual machines (AWS EC)
- PaaS - app development platforms (Google App Engine)
- Saas - ready software (Microsoft 365)

Question 26

Version Control tracks changes in Code

Git is widely used to manage collaboration.

Commit - saved change

Branch - seperate development path

Question 27

Jupyter Notebook is an interactive tool used mainly with python

It is popular for data analysis and visualisation

Question 28

Python is a popular programming language for data and AI

Libraries

- Pandas - data analysis
- NumPy - calculations
- Matplotlib - charts
- SCIKIT-Learn - Machine learning

Section D : Ethics and Careers

Question 29

Data privacy protects personal information
GDPR General Data Protection Regulation is a law
protecting EU citizens data and applies to many countries

Question 30

Data ethics means using data responsibly
Examples of unethical use
1. Data misuse
2. Biased algorithms
3. Privacy violation

Question 31

Data Governance defines rules for managing data.
Key components
1. quality
2. security
3. ownership
4. Compliance

Question 32

Bias happens when AI unfairly favours one group
Example - biased hiring systems.
This can cause discrimination

Question 33

Data Roles Comparison

Roles	Main Focus	Tools
Data Analyst	Reports on insights	Excel, SQL
Data Scientist	Models and prediction	Python, Machine Learning
Data Engineer	Pipelines and Storage	SQL, Spark
AI ~~Data~~ Engineer	AI systems	Machine Learning framework

Question 34

Business Intelligence (BI) uses data to support decisions
Tools Power BI Tableau

Section E : Personal Reflection

Question 35

Insight into how AI works
Like Generative AI because it can create content like human

Question 36

Neural networks were difficult
Used videos and online articles to understand

Question 37

Data and AI will create new career opportunities and help automate work to optimize my outputs.

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